

A.2 Forces and Momentum

Past Paper Questions

These are past paper questions selected from past papers (2016 - 2020) that relate to the topic A.2 Forces and Momentum. Answers can be found in the separate accompanying document

Question 1

An object of mass m rests on a horizontal plane. The angle θ that the plane makes with the horizontal is slowly increased from zero. When $\theta = \theta_0$, the object begins to slide. What are the coefficient of static friction μ_s and the normal reaction force N of the plane at $\theta = \theta_0$?

	μ_s	N
A.	$\sin \theta_0$	$mg \cos \theta_0$
B.	$\tan \theta_0$	$mg \sin \theta_0$
C.	$\sin \theta_0$	$mg \sin \theta_0$
D.	$\tan \theta_0$	$mg \cos \theta_0$

- **Learning Objective:** A.2.6
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2016, Paper 1
- **Question type:** Multiple Choice

Question 2

A spring of negligible mass and length l_0 hangs from a fixed point. When a mass m is attached to the free end of the spring, the length of the spring increases to l . The tension in the spring is equal to $k\Delta x$, where k is a constant and Δx is the extension of the spring. What is k ?

- A. $\frac{mg}{l_0}$
- B. $\frac{mg}{l}$
- C. $\frac{mg}{l - l_0}$
- D. $\frac{mg}{l_0 - l}$

- **Learning Objective:** A.2.6
- **Level:** Standard Level
- **Past Paper:** May 2016, Paper 1
- **Question type:** Multiple Choice

Question 3

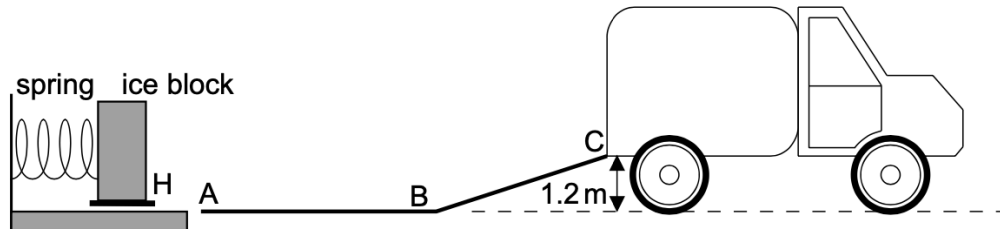
A ball with mass m moves horizontally with speed u . The ball hits a vertical wall and rebounds in the opposite direction with speed $v < u$. The duration of the collision is T . What are the magnitude of the average force exerted by the wall on the ball and the loss of kinetic energy of the ball?

	Average force	Loss of kinetic energy
A.	$\frac{m(u + v)}{T}$	$\frac{m(u^2 - v^2)}{2}$
B.	$\frac{m(u + v)}{T}$	$\frac{m(u - v)^2}{2}$
C.	$\frac{m(u - v)}{T}$	$\frac{m(u^2 - v^2)}{2}$
D.	$\frac{m(u - v)}{T}$	$\frac{m(u - v)^2}{2}$

- **Learning Objective:** A.2.9, A.2.11
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2016, Paper 1
- **Question type:** Multiple Choice

Question 4

A company designs a spring system for loading ice blocks onto a truck. The ice block is placed in a holder H in front of the spring and an electric motor compresses the spring by pushing H to the left. When the spring is released the ice block is accelerated towards a ramp ABC. When the spring is fully decompressed, the ice block loses contact with the spring at A. The mass of the ice block is 55 kg.



Assume that the surface of the ramp is frictionless and that the masses of the spring and the holder are negligible compared to the mass of the ice block.

Describe the motion of the block

- (i) from A to B with reference to Newton's first law. [1]
- (ii) from B to C with reference to Newton's second law. [2]

- **Learning Objective:** A.2.1
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2016, Paper 2
- **Question type:** Explain

Question 5

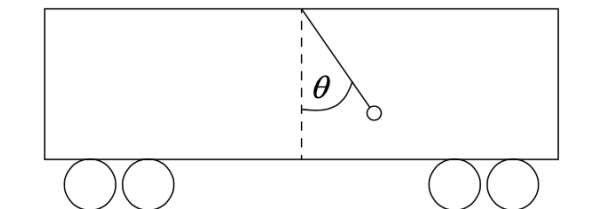
A boy jumps from a wall 3 m high. What is an estimate of the change in momentum of the boy when he lands without rebounding?

- A. $5 \times 10^0 \text{ kg ms}^{-1}$
- B. $5 \times 10^1 \text{ kg ms}^{-1}$
- C. $5 \times 10^2 \text{ kg ms}^{-1}$
- D. $5 \times 10^3 \text{ kg ms}^{-1}$

- **Learning Objective:** A.2.9, A.2.11
- **Level:** Standard Level
- **Past Paper:** Nov 2016, Paper 1
- **Question type:** Multiple Choice

Question 6

A mass is suspended from the ceiling of a train carriage by a string. The string makes an angle θ with the vertical when the train is accelerating along a straight horizontal track.



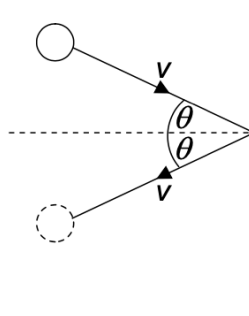
What is the acceleration of the train?

- A. $g \sin \theta$
- B. $g \cos \theta$
- C. $g \tan \theta$
- D. $\frac{g}{\tan \theta}$

- **Learning Objective:** A.2.1
- **Level:** Higher Level
- **Past Paper:** Nov 2016, Paper 1
- **Question type:** Multiple Choice

Question 7

A ball of mass m strikes a vertical wall with a speed v at an angle of θ to the wall. The ball rebounds at the same speed and angle. What is the change in the magnitude of the momentum of the ball?

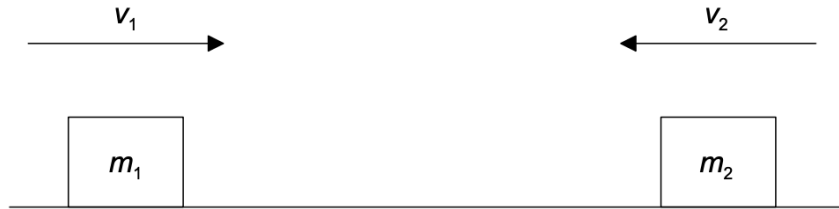


- A. $2mv \sin \theta$
- B. $2mv \cos \theta$
- C. $2mv$
- D. zero

- **Learning Objective:** A.2.8, A.2.11
- **Level:** Standard Level
- **Past Paper:** Nov 2016, Paper 1
- **Question type:** Multiple Choice

Question 8

Two objects m_1 and m_2 approach each other along a straight line with speeds v_1 and v_2 as shown. The objects collide and stick together.



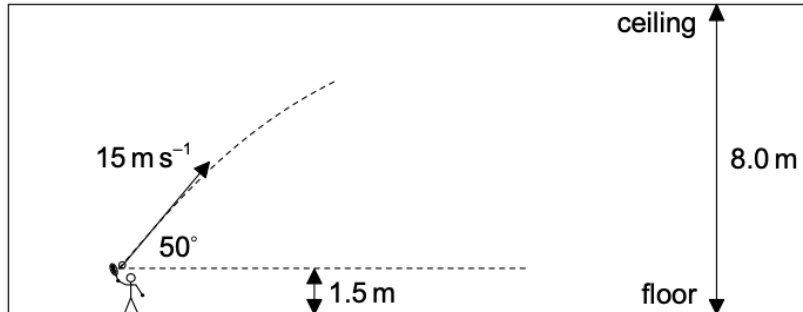
What is the total change of linear momentum of the objects as a result of the collision?

- A. $m_1v_1 + m_2v_2$
- B. $m_1v_1 - m_2v_2$
- C. $m_2v_2 - m_1v_1$
- D. zero

- **Learning Objective:** A.2.9, A.2.11
- **Level:** Standard Level
- **Past Paper:** Nov 2016, Paper 1
- **Question type:** Multiple Choice

Question 9

A tennis ball is hit with a racket from a point 1.5 m above the floor. The ceiling is 8.0 m above the floor. The initial velocity of the ball is 15 m s^{-1} at 50° above the horizontal. Assume that air resistance is negligible.

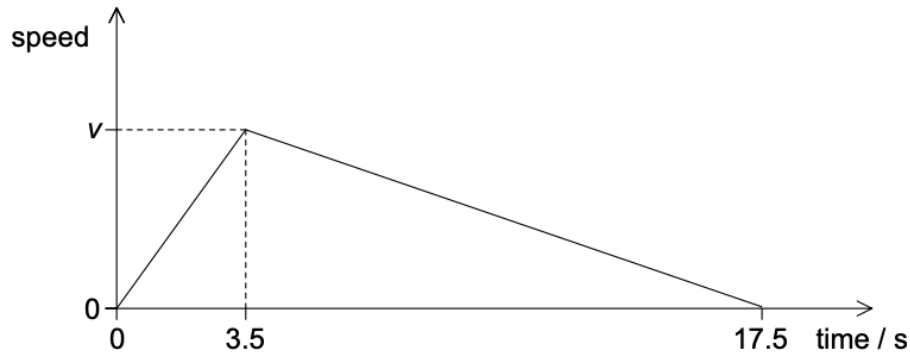


- (a) Determine whether the ball will hit the ceiling. [3]
- (b) The tennis ball was stationary before being hit. It has a mass of $5.8 \times 10^{-2} \text{ kg}$ and was in contact with the racket for 23 ms.
- (i) Calculate the mean force exerted by the racket on the ball. [1]
- (ii) Explain how Newton's third law applies when the racket hits the tennis ball. [2]

- **Learning Objective:** A.2.3, A.2.9, A.2.11
- **Level:** Standard Level and Higher Level
- **Past Paper:** Nov 2016, Paper 2
- **Question type:** Calculate and Explain

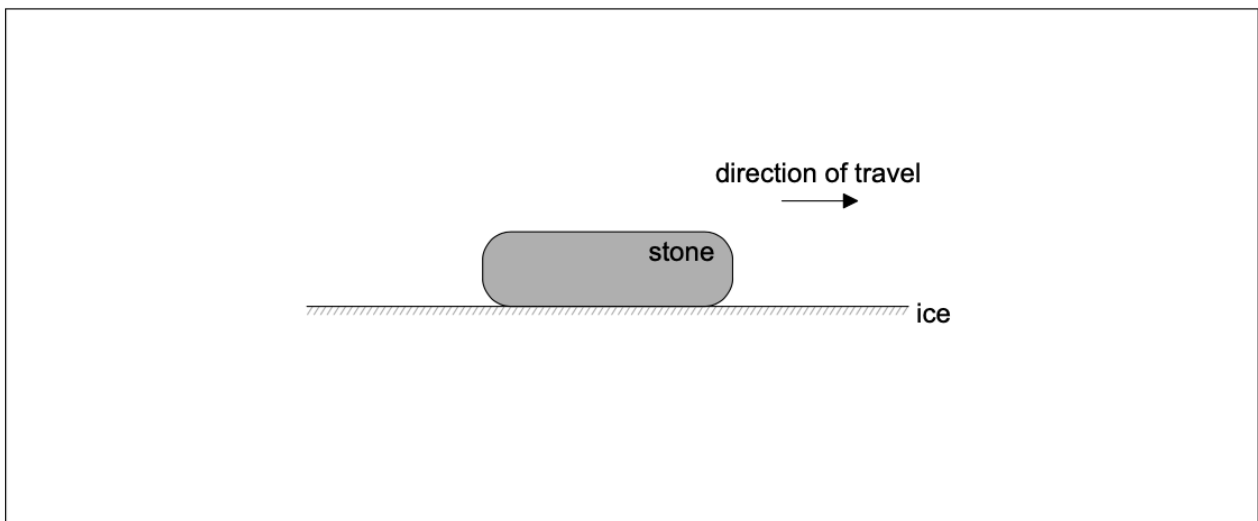
Question 10

Curling is a game played on a horizontal ice surface. A player pushes a large smooth stone across the ice for several seconds and then releases it. The stone moves until friction brings it to rest. The graph shows the variation of speed of the stone with time.



The total distance travelled by the stone in 17.5 s is 29.8 m.

- (a) Determine the maximum speed v of the stone. [2]
- (b) Determine the coefficient of dynamic friction between the stone and the ice during the last 14.0 s of the stone's motion. [3]
- (c) The diagram shows the stone during its motion **after** release.

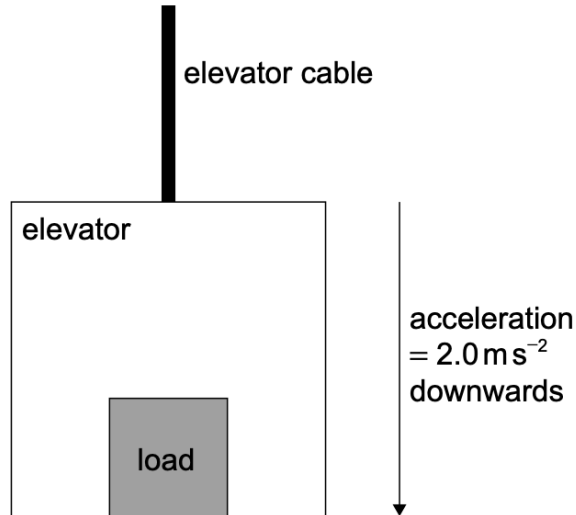


Label the diagram to show the forces acting on the stone. Your answer should include the name, the direction **and** point of application of each force. [3]

- **Learning Objective:** A.2.6, A.2.5
- **Level:** Standard Level and Higher Level
- **Past Paper:** Nov 2016, Paper 2
- **Question type:** Calculate and Draw

Question 11

An elevator (lift) and its load have a total mass of 750 kg and accelerate vertically downwards at 2.0 m s^{-2} .



What is the tension in the elevator cable?

- A. 1.5 kN
- B. 6.0 kN
- C. 7.5 kN
- D. 9.0 kN

- **Learning Objective:** A.2.1
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2017, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 12

An inelastic collision occurs between two bodies in the absence of external forces.

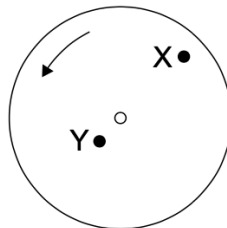
What must be true about the total momentum of the two bodies and the total kinetic energy of the two bodies during this interaction?

- A. Only momentum is conserved.
- B. Only kinetic energy is conserved.
- C. Both momentum and kinetic energy are conserved.
- D. Neither momentum nor kinetic energy are conserved.

- **Learning Objective:** A.1 Math Skill 3
- **Level:** Standard Level
- **Past Paper:** May 2017, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 13

A horizontal disc rotates uniformly at a constant angular velocity about a central axis normal to the plane of the disc.



Point X is a distance $2L$ from the centre of the disc. Point Y is a distance L from the centre of the disc. Point Y has a linear speed v and a centripetal acceleration a .

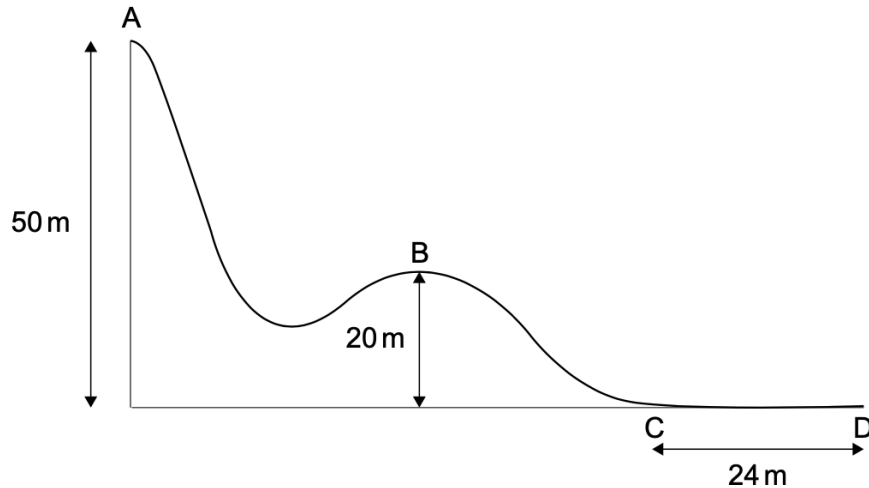
What is the linear speed and centripetal acceleration of point X?

	Linear speed of X	Centripetal acceleration of X
A.	v	a
B.	$2v$	$2a$
C.	v	$2a$
D.	$2v$	$4a$

- **Learning Objective:** A.2.16
- **Level:** Standard Level
- **Past Paper:** May 2017, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 14

The diagram below shows part of a downhill ski course which starts at point A, 50 m above level ground. Point B is 20 m above level ground.



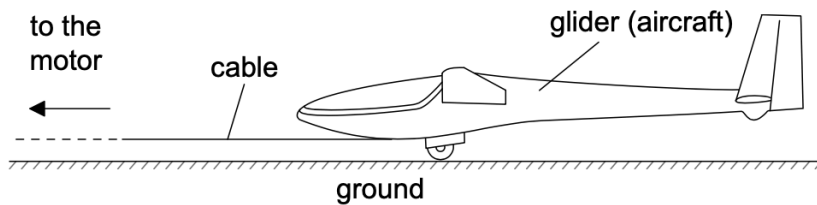
At the side of the course flexible safety nets are used. Another skier of mass 76 kg falls normally into the safety net with speed 9.6 m s^{-1} .

- (i) Calculate the impulse required from the net to stop the skier and state an appropriate unit for your answer. [2]
- (ii) Explain, with reference to change in momentum, why a flexible safety net is less likely to harm the skier than a rigid barrier. [2]

- **Learning Objective:** A.2.9, A.2.11
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2017, Paper 2, TZ1
- **Question type:** Calculate and Explain

Question 15

A glider is an aircraft with no engine. To be launched, a glider is uniformly accelerated from rest by a cable pulled by a motor that exerts a horizontal force on the glider throughout the launch.



The cable is wound onto a cylinder of diameter 1.2 m. Calculate the angular velocity of the cylinder at the instant when the glider has a speed of 27 m s^{-1} . Include an appropriate unit for your answer.

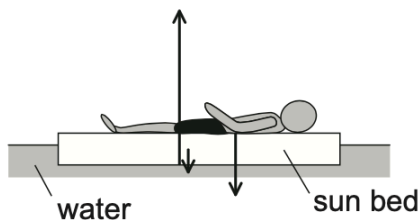
[2]

- **Learning Objective:** A.2.16
- **Level:** Standard Level
- **Past Paper:** May 2017, Paper 2, TZ2
- **Question type:** Calculate

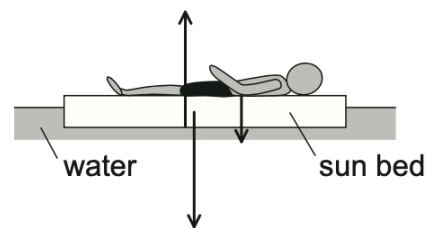
Question 16

A sunbather is supported in water by a floating sun bed. Which diagram represents the magnitudes of the forces acting on the sun bed?

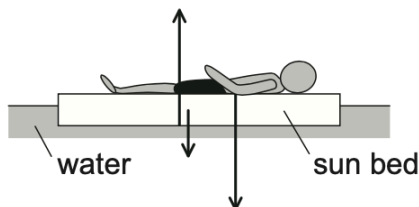
A.



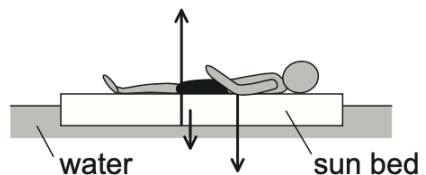
B.



C.



D.



- **Learning Objective:** A.2.6
- **Level:** Higher Level
- **Past Paper:** Nov 2017, Paper 1
- **Question type:** Multiple Choice

Question 17

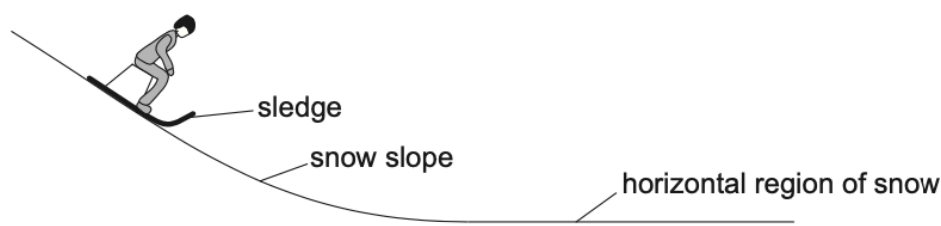
A toy car of mass 0.15 kg accelerates from a speed of 10 cm s^{-1} to a speed of 15 cm s^{-1} . What is the impulse acting on the car?

- A. 7.5 mN s
- B. 37.5 mN s
- C. 0.75 N s
- D. 3.75 N s

- **Learning Objective:** A.2.9, A.2.11
- **Level:** Higher Level
- **Past Paper:** Nov 2017, Paper 1
- **Question type:** Multiple Choice

Question 18

A girl on a sledge is moving down a snow slope at a uniform speed.

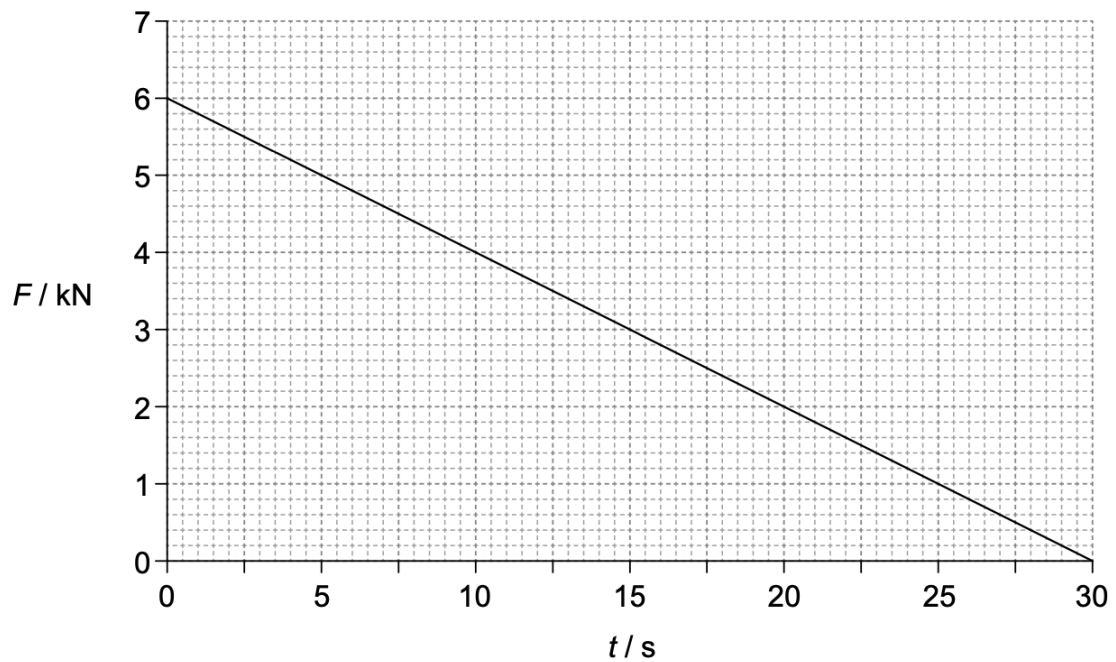


- (a) Draw the free-body diagram for the sledge at the position shown on the snow slope. [2]
- (b) After leaving the snow slope, the girl on the sledge moves over a horizontal region of snow. Explain, with reference to the physical origin of the forces, why the vertical forces on the girl must be in equilibrium as she moves over the horizontal region. [3]
- (c) When the sledge is moving on the horizontal region of the snow, the girl jumps off the sledge. The girl has no horizontal velocity after the jump. The velocity of the sledge immediately after the girl jumps off is 4.2 m s^{-1} . The mass of the girl is 55 kg and the mass of the sledge is 5.5 kg . Calculate the speed of the sledge immediately before the girl jumps from it. [2]
- (d) The girl chooses to jump so that she lands on loosely-packed snow rather than frozen ice. Outline why she chooses to land on the snow. [3]
- (e) The sledge, without the girl on it, now travels up a snow slope that makes an angle of 6.5° to the horizontal. At the start of the slope, the speed of the sledge is 4.2 m s^{-1} . The coefficient of dynamic friction of the sledge on the snow is 0.11 .
 - (i) Show that the acceleration of the sledge is about -2 m s^{-2} . [3]
 - (ii) Calculate the distance along the slope at which the sledge stops moving. Assume that the coefficient of dynamic friction is constant. [2]
- (f) The coefficient of static friction between the sledge and the snow is 0.14 . Outline, with a calculation, the subsequent motion of the sledge. [2]

- **Learning Objective:** A.2.5, A.2.2, A.2.9, A.2.6
- **Level:** Standard Level and Higher Level
- **Past Paper:** Nov 2017, Paper 2
- **Question type:** Draw, Explain, Calculate

Question 19

The graph shows the variation with time t of the force F acting on an object of mass 15 000 kg. The object is at rest at $t=0$.



What is the speed of the object when $t=30 \text{ s}$?

- A. 0.18 m s^{-1}
- B. 6 m s^{-1}
- C. 12 m s^{-1}
- D. 180 m s^{-1}

- **Learning Objective:** A.2.9
- **Level:** Standard and Higher Level
- **Past Paper:** May 2018, Paper 1, TZ2
- **Question type:** Multiple Choice

Question 20

Satellite X orbits a planet with orbital radius R . Satellite Y orbits the same planet with orbital radius $2R$. Satellites X and Y have the same mass.

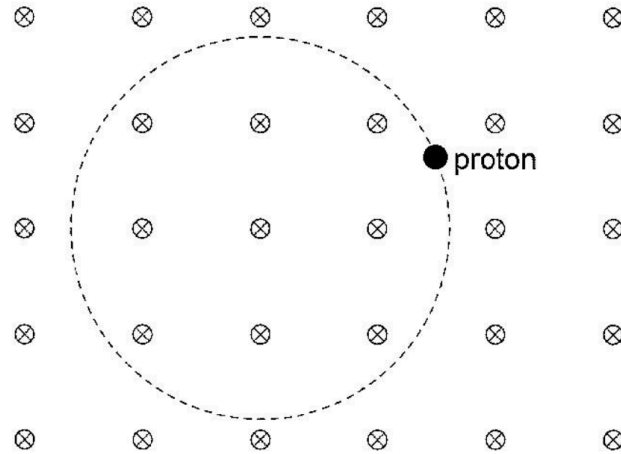
What is the ratio $\frac{\text{centripetal acceleration of X}}{\text{centripetal acceleration of Y}}$?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 2
- D. 4

- **Learning Objective:** A.2.16
- **Level:** Standard Level and Higher Level
- **Past Paper:** May 2019, Paper 1, TZ1
- **Question type:** Multiple Choice

Question 21

A proton moves along a circular path in a region of a uniform magnetic field. The magnetic field is directed into the plane of the page.



(a) Label with arrows on the diagram the

(i) magnetic force F on the proton.

[1]

(ii) velocity vector v of the proton.

[1]

- **Learning Objective:** A.2.15
- **Level:** Standard and Higher Level
- **Past Paper:** May 2019, Paper 2, TZ2
- **Question type:** Draw